Ventilators on the Fly — Clinical One-Pager

Read any vent with 4 questions: mode · O_2 support · ventilation · mechanics. Adapted from TeachIM.org . Educational use only.

1 · What is the mode?

Mode	Square wave	Mand. RR	Target	Monitor
AC-PC	Pressure	Yes	$P_{control}$	VT, VE
AC-VC	Flow	Yes	VT	Peak, Plateau
PRVC	Pressure	Yes	VT	Peak/Plat, achieved VT
PS	Pressure	**No**	$P_{support}$	VT, RR, VE

Square in **pressure** = pressure mode · square in **flow** = volume control. Set RR = control; no set RR = support. (Modern vents can change VC flow shape → absent flow-square doesn't exclude VC.)

2 · Level of O_2 support

FiO_2	30%	50%	70%	100%
PEEP	5	8	10	18–24

FiO_2 + PEEP set oxygenation in every mode. Higher pair on the ARDSNet ladder = more O_2 support / sicker lungs.

3 · Is ventilation appropriate?

- **VE = VT × RR**; normal 4–6 L/min (often higher in disease).
- Judge by the **pH** (goal 7.2–7.5), not the absolute VE.
- Vent- or patient-triggered? Patient tachypnea → CNS injury, pain, salicylates, metabolic acidosis.

4 · Respiratory mechanics

Pressure modes: watch **ΔP** (good lungs need ΔP <10 for ~6 mL/kg; falling VT = trouble). Volume modes: watch **peak & plateau**.
If ΔP > 10 in a volume mode → inspiratory hold for the plateau:

Finding	Problem	Causes
Peak – Plat > 3–5	Resistance	mucus plug, kinked tube, bronchospasm
Plat – PEEP > 10	Compliance	pneumothorax, edema/alveolar filling, fibrosis, mainstem intubation, distended abdomen

Lung-protective targets (ARDS)

- **VT 6 mL/kg** predicted body weight (4–8) — ARDSNet, NEJM 2000.
- **Plateau <30** cm H_2O (many target ≤28).
- **Driving pressure** (Plateau – PEEP) is the variable most linked to survival — target **≤14–15** (Amato, NEJM 2015). Bedside ΔP uses peak; true driving pressure uses the **plateau**.

Orientation & quick rules

Screen: right-side dials = clinician-set; left-side numbers = ventilator-measured (can differ if patient breathes over set rate).
Waveforms: PEEP = pressure after exhalation; Peak = max insp pressure; ΔP = Peak – PEEP. Flow up = in, down = out. Expiration is passive → looks the same across all modes.
PS caveat: no mandatory RR → apnea triggers a backup control mode.